

Notes: Da, Fang, and Lin (RFS, 2025)

Fractional Trading

1 Summary

Da, Fang, and Lin study how fractional trading (FT) changes retail participation and price dynamics in high-priced stocks. Fractional trading lets investors trade less than one whole share, so a small-account investor no longer needs hundreds or thousands of dollars to obtain exposure to a high nominal-price stock.

The paper's central empirical idea is that FT should matter most for high-priced stocks because the whole-share affordability constraint binds most strongly there. The authors use off-exchange one-share trades as a proxy for tiny trades generated by fractional-trading orders. They then compare high-priced and low-priced stocks before and after major brokers introduce FT.

The main findings are threefold. First, after FT becomes available, high-priced stocks experience a much larger increase in off-exchange one-share trades than comparable low-priced stocks. Second, these tiny trades matter when retail attention is coordinated: high-priced stocks receiving attention shocks experience more tiny trades, more social media discussion, stronger temporary price pressure, and subsequent price reversals. Third, FT can help generate meme-stock-like trading frenzies and bubbles in high-priced stocks, especially when social media coordination and feedback effects are likely to be present.

2 Research Question

The paper asks:

Does removing the whole-share trading constraint through fractional trading increase retail activity in high-priced stocks, and can the resulting tiny trades affect prices?

The paper is not simply about whether more retail investors enter the stock market. It is about a specific market-design change: investors who previously could not buy a whole share of a high-priced stock can now buy a slice. This changes the effective investor base of high nominal-price stocks and may alter the price impact of coordinated retail attention.

3 Institutional Setting

FT is introduced by major U.S. brokers in late 2019 and 2020. Interactive Brokers announces FT on November 25, 2019. Robinhood follows on December 12, 2019. Fidelity introduces FT on January 29, 2020. Charles Schwab introduces a more limited FT service on June 9, 2020, restricted to S&P 500 stocks.

The timing is useful because FT is introduced after the major move to zero-commission trading. This allows the authors to separate FT from zero-commission trading. The staggered broker-level timing also lets the paper distinguish a modest initial effect after Interactive Brokers and Robinhood from a much larger effect after Fidelity, a major retail broker, enters.

The economic intuition is straightforward. Suppose an investor has \$180 and wants to buy a stock trading at \$100. Without FT, the investor can buy one share and has unused cash. With FT, the investor can buy roughly 1.8 shares. If the stock trades at \$1,000, whole-share trading may prevent the investor from entering at all, while FT makes entry possible with a small dollar order.

4 Measurement: Off-Exchange One-Share Trades

The main empirical proxy for FT-facilitated retail activity is the number of off-exchange one-share trades. The measurement logic comes from the reporting mechanics of fractional orders. At many FT brokers, the whole-share portion of a fractional order is executed on an agency basis, while the fractional-share portion is executed on a principal basis and reported to FINRA trade reporting facilities as a one-share trade under the rounding-up rule.

Therefore, a surge in off-exchange one-share trades is informative about tiny retail trading. The measure may capture both direct fractional trades and the indirect effect of FT: after a small investor becomes comfortable entering high-priced stocks, the investor may also place very small whole-share trades.

This proxy is especially useful because actual fractional-share trades are not fully observable in public TAQ data for the whole sample. The paper later uses a direct fractional-trade measure from Bartlett, McCrary, and O’Hara (2023b) as a robustness check in the stock-bubble analysis.

5 Baseline Difference-in-Differences Design

The baseline sample runs from January 2019 to December 2020. Stocks are classified as high-priced if their nominal share price is at least \$100 at the end of November 2019. Low-priced stocks are below that cutoff. The paper also uses \$75 and \$150 cutoffs as robustness checks.

The baseline OLS difference-in-differences specification is:

$$\#OneShareTrades_{it} = \alpha + \beta_1 HighPrice_i \times PostIBFID_t + \beta_2 HighPrice_i \times PostFID_t + \gamma' X_{it} + \mu_i + \tau_t + \varepsilon_{it}. \quad (1)$$

Here $PostIBFID_t$ covers the period when FT is available through Interactive Brokers or Robinhood but not yet through Fidelity. $PostFID_t$ covers the period after Fidelity introduces FT. Firm fixed effects absorb time-invariant stock differences, date fixed effects absorb market-wide variation in retail trading, and controls include size, book-to-market, earnings-announcement days, past returns, and volatility.

The identifying comparison is whether high-priced stocks experience a larger increase in tiny trades than matched low-priced stocks exactly when the trading constraint is relaxed.

6 Main Result: FT Increases Tiny Trades in High-Priced Stocks

The baseline results show that high-priced stocks experience a relative increase of about 14.1% in daily off-exchange one-share trades after the first FT introduction period, and a much larger increase

of about 138.7% after Fidelity introduces FT. The effect is concentrated among low-institutional-ownership stocks, which is consistent with the interpretation that the activity is retail-driven.

The result survives multiple checks. It remains under alternative price cutoffs. It is not explained by zero-commission trading, because the effect appears after FT timing rather than in the zero-commission window. It is not purely a COVID-19 effect, because the FT effect starts before the pandemic shock becomes central, although the pandemic period may amplify retail activity.

7 Additional Identification Evidence

The paper adds two broker-specific analyses. First, using Robinhood stock-level ownership data, it shows that after Robinhood introduces FT, high-priced stocks experience a larger increase in Robinhood trading intensity than low-priced stocks. This is useful because Robinhood users are a direct retail-investor population.

Second, using Charles Schwab’s partial FT offering for S&P 500 stocks, the paper compares high-priced S&P 500 stocks with high-priced non-S&P 500 stocks and with low-priced S&P 500 stocks around Schwab’s FT launch. The limited Schwab rollout creates a sharper treatment comparison because only S&P 500 stocks are eligible for Schwab’s FT service.

Together, these tests support the causal interpretation that FT, rather than a general retail-trading trend, increases tiny trades in high-priced stocks.

8 Price Pressure During Retail-Attention Events

After documenting trading effects, the paper asks whether tiny trades are economically meaningful. The authors construct a daily retail-attention “super set” of stocks using top winners, top losers, and stocks with large abnormal Google search volume. The idea is that attention makes retail investors coordinate on the same stocks.

The first price-pressure specification is:

$$\#OneShareTrades_{i,t+1} = \alpha + \beta_1 HighPrice_{it} \times PostFT_{t+1} + \gamma' X_{it} + \mu_i + \tau_t + \varepsilon_{it}. \quad (2)$$

The same event framework is then used for social-media discussion and for subsequent return reversals. The return-reversal specification is:

$$BHAR_{i,[t+2,t+6]} = \alpha + \beta_1 HighPrice_{it} \times PostFT_{t+1} + \gamma' X_{it} + \mu_i + \tau_t + \varepsilon_{it}. \quad (3)$$

The results show that after FT, high-priced attention stocks experience more off-exchange one-share trades and more Reddit WallStreetBets discussion than low-priced attention stocks. Their subsequent five-day buy-and-hold abnormal returns are lower, consistent with temporary price pressure and reversal.

9 Meme-Stock-Like Trading Frenzies and Bubbles

The paper then studies whether FT contributes to stock bubbles. A stock is classified as experiencing a bubble if, over a maximum six-month window, its price first rises by more than 150% and

then falls by at least 40% from the peak.

The monthly bubble specification is:

$$\Pr(\text{Bubble}_{im} = 1) = \alpha + \beta_1 \text{HighPrice}_{im} + \beta_2 \text{PostFT}_m + \beta_3 \text{HighPrice}_{im} \times \text{PostFT}_m + \gamma' X_{im} + \varepsilon_{im}. \quad (4)$$

The interaction coefficient is positive: high-priced stocks become more likely to experience bubble episodes after FT. This result is stronger among low-IO stocks and robust to alternative price cutoffs and an alternative pre-FT period based on the dot-com era.

For bubble events, the paper estimates:

$$R_{it} = \alpha + \beta_1 \% \text{TinyTrades}_{it} + \beta_2 \text{HighPrice}_{im} \times \% \text{TinyTrades}_{it} + \gamma' X_{it} + \varepsilon_{it}. \quad (5)$$

The interaction is positive. During bubble events, tiny trades track daily returns more strongly for high-priced stocks. Using direct fractional-trade measures delivers similar evidence.

10 Feedback Effect Interpretation

The final mechanism tests connect to feedback models. If price increases attract investor attention and capital, then coordinated retail trading can create a reinforcing loop: retail buying pushes prices up, price increases attract more attention and trading, and the loop can continue until the bubble reverses.

The paper tests whether the FT-bubble relation is stronger where feedback effects should be more important: stocks with more WallStreetBets discussion, stocks with lower credit ratings, and stocks with higher estimated capital-to-price sensitivity. The triple-interaction tests support this interpretation.

The result is important because it suggests FT is not only a mechanical access shock. It can interact with social coordination and financing feedback to amplify price dynamics.

11 Interpretation

The paper reframes nominal share price. Before FT, a high nominal price can mechanically limit small-account retail participation. After FT, high nominal price no longer prevents entry. This can be welfare-improving if FT broadens access and democratizes investment opportunities. But it can also increase the price impact of coordinated retail noise trading.

The main economic tension is therefore access versus fragility. FT lowers participation barriers, but by enabling many small investors to coordinate in high-priced stocks, it can also make those stocks more exposed to retail attention shocks, temporary price pressure, and meme-style bubbles.

12 Connection to Broader Literature

The paper connects to several literatures. In market microstructure, it provides evidence that tiny retail trades can move prices when they are coordinated. In behavioral finance, it links attention,

social media, and nominal-price constraints to retail trading. In corporate finance and market design, it shows that trading technology can alter the effective investor base and change asset-pricing outcomes.

It also speaks to the nominal share price literature. If investors suffer from nominal price illusion or think in share units rather than dollar units, then high-priced stocks used to be less attractive to small investors. FT partly breaks that link by allowing dollar-based fractional orders.

13 Main Takeaways

1. **FT removes a binding trading constraint.** Small investors can enter high-priced stocks without buying a whole share.
2. **Off-exchange one-share trades proxy for FT-facilitated tiny trading.** The measure is tied to the reporting mechanics of fractional orders.
3. **High-priced stocks receive the largest trading response.** Tiny trades rise much more for high-priced stocks after FT is introduced, especially after Fidelity's entry.
4. **The effect is retail-driven.** It is stronger among low-IO stocks and is validated using Robinhood and Charles Schwab tests.
5. **Tiny trades can create price pressure.** During retail-attention events, high-priced stocks experience more tiny trades, more social-media discussion, and subsequent return reversals.
6. **FT can fuel meme-stock-like bubbles.** After FT, high-priced stocks become more likely to experience bubble events, especially when feedback channels are strong.

14 Original Figures and Tables from the Paper

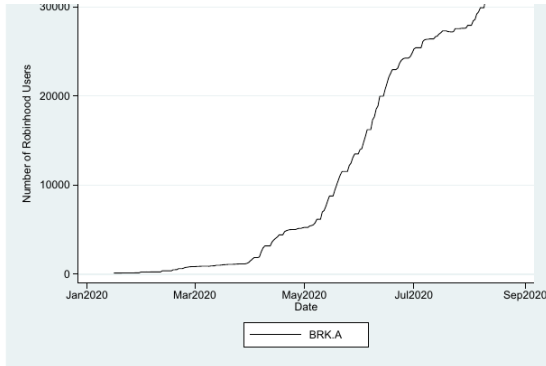


Figure 1
Number of Robinhood users holding BRK.A
This figure plots the daily number of Robinhood users holding BRK.A during the sample period from January 16, 2020, to August 13, 2020. BRK.A has the highest nominal share price among the listed stocks.

Figure 1: Robinhood users holding BRK.A.

Table 1
Summary statistics

A. Prematch sample						
Variable	Obs	Mean	SD	P25	P50	P75
Price	2,385	57.587	70.265	16.570	34.360	70.240
# of one-share trades _t	1,192,500	0.153	0.408	0.012	0.044	0.125
ln(Market cap) _{t-1}	1,192,500	14.351	1.843	12.980	14.276	15.531
Book to market _{t-1}	1,192,500	0.679	0.326	0.398	0.700	0.963
Past month volatility _t	1,192,500	2.934	2.082	1.580	2.318	3.591
Past month max return _t	1,192,500	0.063	0.054	0.029	0.046	0.077
Past week return _t	1,192,500	0.004	0.073	-0.027	0.003	0.034
Past month return _t	1,192,500	0.020	0.150	-0.050	0.017	0.087
Past year return _t	1,192,500	0.024	0.420	-0.232	-0.032	0.195

This panel reports summary statistics of the continuous variables used in the baseline analyses linking FT introduction to small retail activity for the pre-match sample and the stock price. The sample period is from January 2, 2019, to December 31, 2020. Price is the firm's nominal stock price at the end of November 2019. # of one-share trades_t is the number of off-exchange one-share trades on day *t* in thousands. ln(Market cap)_{t-1} is the log of market capitalization on the previous trading day. Book to market_{t-1} is the book-to-market of prior quarter. Past month volatility_t is the standard deviation of the stock's daily returns over the past month in percentage points. Past month max return_t is the stock's maximum daily return over the past month. Past week return_t, Past month return_t, and Past year return_t are the stock's cumulative return of the past week, month, and year, respectively. Detailed variable definitions are in Appendix. All variables are winsorized at the top and bottom 1% by trading days.

Table 1A: Prematch summary statistics.

Read: Figure 1 motivates the paper with BRK.A, an extremely high-priced stock that nevertheless becomes accessible to Robinhood users. Table 1A describes the prematch sample used in the baseline trading analysis.

Economic message: FT turns nominally inaccessible stocks into investable stocks for small accounts, so high nominal prices become a treatment-relevant friction.

Table 1
(Continued)

B. Postmatch firm characteristics

	Obs	Low-priced	High-priced	p-value
Pooled (\$100) sample				
Book to market _{t-1}	708	0.446	0.460	.48
Popularity	708	5,451.404	3,729.749	.29
Institutional ownership	708	0.822	0.828	.61
Growth of # of one-share trades _t	708	1.072	1.044	.96
Low-IO subsample				
Book to market _{t-1}	282	0.526	0.544	.60
Popularity	282	5,354.142	7,045.355	.55
Institutional ownership	282	0.686	0.703	.30
Growth of # of one-share trades _t	282	0.883	1.511	.46
High-IO subsample				
Book to market _{t-1}	342	0.423	0.434	.67
Popularity	342	1,512.374	1,277.678	.60
Institutional ownership	342	0.936	0.937	.83
Growth of # of one-share trades _t	342	1.083	1.206	.69
Pooled (\$75) sample				
Book to market _{t-1}	850	0.555	0.541	.47
Popularity	850	2,868.012	2,999.72	.90
Institutional ownership	850	0.829	0.823	.56
Growth of # of one-share trades _t	850	0.990	1.029	.92
Pooled (\$150) sample				
Book to market _{t-1}	360	0.408	0.400	.77
Popularity	360	5,144.211	4,196.578	.68
Institutional ownership	360	0.824	0.820	.82
Growth of # of one-share trades _t	360	1.799	1.549	.73

This panel reports summary statistics of the high- and low-priced groups and the differences between the two groups within the three pooled samples (based on different price cutoffs) as well as the two subsamples of low- and high-IO stocks (based on the first price cutoff) after PSM. The pooled sample, from January 2, 2019, to December 31, 2020, includes observations pulled from both groups that are matched on Fama-French industry, book-to-market, stock popularity, institutional ownership, and growth in the daily number of off-exchange one-share trades pre-FT. The high- (low-) priced group includes firm-year observations with nominal share price of \$100, \$75, and \$150 or above (below \$100, \$75, and \$150) at the end of November 2019. The high- and low-IO subsamples are divided within high- and low-priced groups based on how a stock's institutional ownership compares to the group median at the end of November 2019. The first column of each subpanel reports the number of firms for the subsample after PSM. The second and third columns of each subpanel report the mean of variables in low and high-priced groups, respectively. The last column of each subpanel reports the *p*-value of the two-tailed *t*-test. Book to market_{t-1} is the book-to-market of prior year. Popularity is the number of Robinhood users holding a given stock at the end of November 2019. Institutional ownership is the percentage of institutional investors holding a given stock at the end of November 2019. The growth measure for # of one-share trades_t variable is calculated as the cumulative daily values over the 5-month period of June-October 2019 minus the cumulative daily values over the 5-month period of January-May 2019. Detailed variable definitions are in Appendix.

Table 1B: Postmatch firm characteristics.

Table 1
(Continued)

C. Postmatch sample summary

Variable	Obs	Mean	SD	P25	P50	P75
Price	708	129.652	172.967	39.445	98.875	150.930
# of one-share trades _t	354,000	0.274	0.633	0.039	0.098	0.237
ln(Market cap) _{t-1}	354,000	15.463	1.718	14.256	15.430	16.624
Book to market _{t-1}	354,000	0.462	0.275	0.239	0.399	0.638
Past month volatility _t	354,000	2.634	1.812	1.466	2.107	3.174
Past month max return _t	354,000	0.056	0.047	0.027	0.041	0.068
Past week return _t	354,000	0.006	0.065	-0.023	0.005	0.033
Past month return _t	354,000	0.024	0.135	-0.040	0.023	0.087
Past year return _t	354,000	0.122	0.423	-0.135	0.068	0.299

This panel reports summary statistics of the continuous variables used in the baseline analyses linking FT introduction to small retail activity and the stock price for the pooled sample after PSM. The pooled sample, from January 2, 2019, to December 31, 2020, includes observations pulled from both groups that are matched on Fama-French industry, book-to-market, stock popularity, institutional ownership, and growth in the daily number of off-exchange one-share trades pre-FT. The high- (low-) priced group includes firm-day observations with nominal share price of \$100 or above (below \$100) at the end of November 2019. Variables are as in Table 1, panel A. Detailed variable definitions are in Appendix. All variables are winsorized at the top and bottom 1% by trading days.

Table 1C: Postmatch sample summary.

Read: Panels B and C show the postmatching balance and summary statistics for the high- and

low-priced comparison groups.

Economic message: The empirical design tries to compare high-priced and low-priced stocks that were similar on observables before FT.

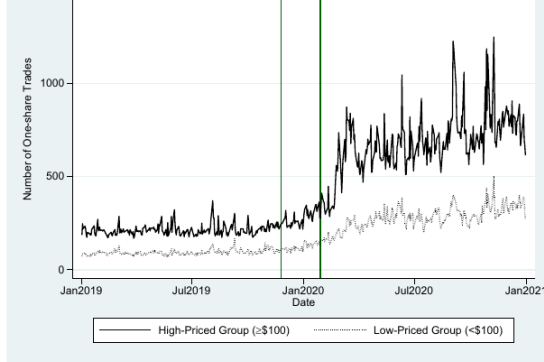


Figure 2
Number of off-exchange one-share trades for high- versus low-priced groups
This figure plots the number of off-exchange one-share trades for high- versus low-priced groups. The sample period is from January 2, 2019, to December 31, 2020. The solid line indicates the high-priced group (i.e., stocks with nominal share price of \$100 or above at the end of November 2019) and the dotted line represents the low-priced group (i.e., stocks with nominal share price below \$100 at the end of November 2019), respectively. The two vertical lines represent November 25, 2019 (the day Interactive Brokers introduced FT), and January 29, 2020 (the day Fidelity introduced FT), respectively.

Figure 2: Off-exchange one-share trades.

Table 2
FT and retail trading: Baseline analyses

	# of one-share trades _{<i>t</i>}				
	(1) Pooled (\$100)	(2) Low-IO	(3) High-IO	(4) \$75	(5) \$150
<i>High price × Post-IB-FID_t</i>	0.020** (2.39)	0.050** (2.15)	-0.006 (-0.89)	0.013** (2.29)	0.034** (2.35)
<i>High price × Post-FID_t</i>	0.197*** (4.33)	0.400*** (2.93)	0.056** (2.52)	0.171*** (5.88)	0.235*** (3.41)
<i>lnMarket cap_{t-1}</i>	0.151*** (2.78)	0.138 (0.71)	0.165*** (3.98)	0.092*** (2.68)	0.216** (2.28)
<i>Book to market_{t-1}</i>	0.172 (1.32)	0.294 (0.68)	0.039 (0.65)	-0.019 (-0.29)	0.165 (1.08)
<i>Earnings announcement_t</i>	0.093*** (11.36)	0.137*** (7.21)	0.063*** (11.88)	0.068*** (12.57)	0.117*** (8.77)
<i>Past month volatility_t</i>	0.028*** (4.52)	0.036* (1.71)	0.030*** (5.56)	0.026*** (4.31)	0.037*** (3.69)
<i>Past month max return_t</i>	-0.296*** (-2.75)	-0.093 (-0.20)	-0.225*** (-2.75)	-0.134 (-1.48)	-0.387* (-1.92)
<i>Past week return_t</i>	0.005 (0.17)	0.057 (0.90)	-0.037** (-2.46)	0.029 (1.50)	-0.032 (-0.84)
<i>Past month return_t</i>	-0.059** (-1.98)	-0.182* (-1.67)	-0.048 (-1.58)	-0.036 (-1.48)	-0.052 (-0.99)
<i>Past year return_t</i>	0.079** (2.06)	0.220* (1.86)	-0.008 (-0.42)	0.042** (2.40)	0.044 (1.21)
Observations	354,000	141,000	171,000	425,000	180,000
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	.658	.602	.595	.653	.649

This table reports the ordinary least squares (OLS) regression results on differences in high- and low-priced firms' number of off-exchange one-share trades surrounding the introduction of FT. A firm-day observation is classified into the high-priced group if the firm's nominal stock price equals or exceeds \$100 for columns 1-3 (\$75 for column 4 and \$150 for column 5) at the end of November 2019 and into the low-priced group otherwise. The pooled sample, from January 2, 2019, to December 31, 2020, includes observations pulled from both groups that are matched on Fama-French industry, book-to-market, stock popularity, institutional ownership, and growth in the daily number of off-exchange one-share trades pre-FT. The two subsamples in columns 2 and 3 are divided within high- and low-priced groups based on how a stock's institutional ownership compares to the group median at the end of November 2019 and then matched applying the same PSM procedure. Column 1 uses the pooled sample; column 2 uses the subsample of low-IO stocks; column 3 uses the subsample of high-IO stocks; and columns 4 and 5 use different price cutoffs, respectively. # of one-share trades_{*t*} measures the daily number of off-exchange one-share trades. *High price* indicates whether the firm is in the high-priced group. *Post-IB-FID_t* indicates whether trading day *t* falls in the period when FT was available through either Interactive Brokers or Robinhood, but not yet through Fidelity. *Post-FID_t* indicates whether trading day *t* falls in the period after FT became available through all three brokers. The DiD estimators are *High price × Post-IB-FID_t* and *High price × Post-FID_t*. Controls include the log of market cap on the previous trading day (*lnMarket cap_{t-1}*), book-to-market of prior quarter (*Book to market_{t-1}*), an indicator for quarterly earnings announcement (*Earnings announcement_t*), stock price volatility of past month (*Past month volatility_t*), the stock's maximum daily return of past month (*Past month max return_t*), and the stock's cumulative return of past week, month, and year (*Past week return_t*, *Past month return_t*, and *Past year return_t*) as well as firm and date fixed effects. Detailed variable definitions are in Appendix. Standard errors are clustered by firm and date. All continuous variables are winsorized at the top and bottom 1% by trading days. *t*-statistics are in parentheses. **p* < .1; ***p* < .05; ****p* < .01.

Table 2: Baseline FT effect on tiny trades.

Read: Figure 2 shows the visual divergence between high- and low-priced stocks after FT. Table 2 confirms the divergence in a controlled DiD specification.

Economic message: The trading response is not just a general retail-trading boom; it is disproportionately strong exactly where whole-share constraints should bind.

Table 3
FT and retail trading: Additional analyses

A. Effect of ZCT

	# of one-share trades _{<i>t</i>}				
	(1) Pooled (\$100)	(2) Low-IO	(3) High-IO	(4) \$75	(5) \$150
<i>High price</i> × <i>Post-ZCT-FT_t</i>	-0.012** (-2.31)	-0.008 (-0.63)	-0.016** (-2.45)	-0.016*** (-3.47)	-0.009 (-1.13)
<i>High price</i> × <i>Post-IB-FID_t</i>	0.017** (1.97)	0.048* (1.91)	-0.008 (-1.15)	0.010 (1.63)	0.033** (2.08)
<i>High price</i> × <i>Post-FID_t</i>	0.194*** (4.26)	0.398*** (2.90)	0.053** (2.37)	0.168*** (5.77)	0.234*** (3.35)
Observations	354,000	141,000	171,000	425,000	180,000
Controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	.658	.602	.595	.653	.649

B. Effect of the COVID-19 pandemic

	# of one-share trades _{<i>t</i>}				
	(1) Pooled (\$100)	(2) Low-IO	(3) High-IO	(4) \$75	(5) \$150
<i>High price</i> × <i>Post-IB-FID_t</i>	0.020** (2.40)	0.050** (2.15)	-0.006 (-0.89)	0.013** (2.29)	0.034** (2.35)
<i>High price</i> × <i>Post-FID-COVID_t</i>	0.069*** (2.96)	0.170*** (2.67)	0.014 (1.22)	0.058*** (3.19)	0.106*** (2.79)
<i>High price</i> × <i>Post-COVID_t</i>	0.210*** (4.34)	0.424*** (2.92)	0.060** (2.53)	0.183*** (5.89)	0.249*** (3.40)
Observations	354,000	141,000	171,000	425,000	180,000
Controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	.658	.602	.596	.654	.649

Panel A reports the OLS regression results on differences in high- and low-priced firms' number of off-exchange one-share trades surrounding the introduction of ZCT and FT. Panel B reports the OLS regression results on differences in high- and low-priced firms' number of off-exchange one-share trades surrounding the introduction of FT and COVID-19 pandemic. Column 1 uses the pooled sample; column 2 uses the subsample of low-IO stocks; column 3 uses the subsample of high-IO stocks; and columns 4 and 5 use \$75 and \$150 as price cutoffs for high- and low-priced groups, respectively. Samples are defined the same as in Table 2. # of one-share trades_{*t*} measures the daily number of off-exchange one-share trades. *High price* indicates whether the firm is in the high-priced group. In panel A, *Post-ZCT-FT_t* indicates whether trading day *t* falls in the period when ZCT was available but FT was not. *Post-IB-FID_t* indicates whether trading day *t* falls in the period when FT was available through either Interactive Brokers or Robinhood, but not yet through Fidelity. *Post-FID_t* indicates whether trading day *t* falls in the period after FT became available through all three brokers. The DiD estimators are *High price* × *Post-ZCT-FT_t*, *High price* × *Post-IB-FID_t*, and *High price* × *Post-FID_t*. In panel B, *Post-FID-COVID_t* indicates whether trading day *t* falls in the period after FT was widely available but before the pandemic started. *Post-COVID_t* indicates whether trading day *t* falls in the period after the pandemic started. The DiD estimators are *High price* × *Post-IB-FID_t*, *High price* × *Post-FID-COVID_t*, and *High price* × *Post-COVID_t*. Controls and fixed effects are as in Table 2. Detailed variable definitions are in Appendix. Standard errors are clustered by firm and date. All continuous variables are winsorized at the top and bottom 1% by trading days. *t*-statistics are

Table 4
FT and retail trading: Exchange-specific analyses

A. Evidence from Robinhood

	RH trading intensity _{<i>t</i>}				
	(1) Pooled (\$100)	(2) Low-IO	(3) High-IO	(4) \$75	(5) \$150
<i>High price</i> × <i>Post-RH_t</i>	0.002*** (3.18)	0.002** (2.58)	0.001 (1.56)	0.002*** (5.05)	0.003*** (3.59)
Observations	265,990	106,396	131,010	377,150	134,980
Controls	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	.265	.221	.231	.242	.268

B. Evidence from Charles Schwab

	# of one-share trades _{<i>t</i>}			
	(1) S&P 500	(2) Non-S&P 500	(3) High-Priced	(4) Low-Priced
<i>High price</i> × <i>Post-CS_t</i>	0.167* (1.93)	-0.015 (-1.16)		
<i>S&P 500</i> × <i>Post-CS_t</i>			0.218* (1.91)	-0.001 (-0.01)
Observations	6,230	26,418	5,180	27,468
Controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes
Adjusted R ²	.895	.857	.910	.917

Panel A reports the OLS regression results on differences in high- and low-priced firms' Robinhood trading intensity surrounding the broker's introduction of FT. Panel B reports the OLS regression results on differences in off-exchange one-share trades between high- and low-priced S&P 500 firms, high- and low-priced non-S&P 500 firms, high-priced S&P 500 and non-S&P 500 firms, and low-priced S&P 500 and non-S&P 500 firms surrounding Charles Schwab's FT introduction. In panel A, the classification of firm-day observation into the high- versus low-priced groups and the construction of the low-IO and high-IO subsamples are both as described in Table 2. The pooled sample, from January 2, 2019, to August 13, 2020, includes observations pulled from both groups that are matched on Fama-French industry, book-to-market, stock popularity, and growth in the daily intraday Robinhood retail trading pre-FT. Column 1 uses the pooled sample; column 2 uses the subsample of low-IO stocks; column 3 uses the subsample of high-IO stocks; and columns 4 and 5 use \$75 and \$150 as price cutoffs, respectively. *RH trading intensity_t* measures Robinhood users' trading intensity of a stock during trading day *t*. *High price* indicates whether the firm is in the high-priced group. *Post-RH_t* indicates whether trading day *t* falls in the period after Robinhood began FT. The DiD estimator is *High price* × *Post-RH_t*. In panel B, a firm-day observation is classified into the high-priced group if the firm's nominal stock price as of June 8, 2020, the day before Charles Schwab began FT for the S&P 500 firms, equals or exceeds \$100, and into the low-priced group otherwise. The sample period is from May 29, 2020 (7 trading days before Charles Schwab introduced FT) to June 17, 2020 (7 trading days after Charles Schwab introduced FT). # of one-share trades_{*t*} measures the number of one-share trades on trading day *t*. *High price* × *Post-CS_t* indicates whether the firm is in the high-priced group. *S&P 500* indicates whether the firm is in the S&P 500 index. *Post-CS_t* indicates whether trading day *t* falls in the period when FT was available for S&P 500 stocks through Charles Schwab. The DiD estimators are *Post-CS_t* × *High price* × *Post-CS_t* and *S&P 500* × *Post-CS_t*. In both panels, controls and fixed effects are as in Table 2. Detailed variable definitions are in Appendix. Standard errors are clustered by firm and date. All continuous variables are winsorized at the top and bottom 1% by trading days. *t*-statistics are in parentheses. **p* < .1; ***p* < .05; ****p* < .01.

Table 3: Zero-commission trading and COVID checks.

Table 4: Robinhood and Charles Schwab evidence.

Read: Table 3 separates FT from zero-commission trading and COVID-period effects. Table 4 uses broker-specific evidence from Robinhood and the partial Charles Schwab rollout.

Economic message: The paper strengthens identification by showing that the high-price trading effect lines up with FT availability across multiple institutional tests.

Table 5
FT and price pressure: Retail herding
A. Tiny trades

	# of one-share trades _{t+1}				
	(1) Pooled (\$100)	(2) Low-IO	(3) High-IO	(4) \$75	(5) \$150
<i>High price_t</i>	-0.004 (-0.04)	0.075 (0.20)	-0.046 (-1.54)	-0.045 (-0.67)	0.062 (0.47)
<i>High price_t × Post-FT_{t+1}</i>	0.430*** (5.30)	1.043*** (3.83)	0.149*** (4.19)	0.361*** (5.50)	0.548*** (4.32)
<i>ln(Market cap)_{t-1}</i>	0.229*** (5.24)	0.278*** (3.88)	0.111*** (4.36)	0.229*** (5.23)	0.236*** (5.43)
<i>Book to market_{t-1}</i>	0.104 (1.00)	0.191 (1.29)	0.032 (0.49)	0.102 (0.98)	0.107 (1.04)
<i>Earnings announcement_t</i>	0.068*** (5.66)	0.081*** (3.18)	0.063*** (8.16)	0.068*** (5.65)	0.069*** (5.72)
<i>Past month volatility_t</i>	0.059*** (6.92)	0.057*** (4.26)	0.039*** (7.74)	0.059*** (6.95)	0.058*** (6.91)
<i>Past month max return_t</i>	-0.947*** (-4.55)	-1.193*** (-3.58)	-0.422*** (-3.88)	-0.952*** (-4.59)	-0.948*** (-4.56)
<i>Past week return_t</i>	0.359*** (6.76)	0.550*** (6.49)	0.022 (0.70)	0.360*** (6.79)	0.358*** (6.75)
<i>Past month return_t</i>	0.012 (0.31)	0.027 (0.46)	0.032 (1.30)	0.012 (0.31)	0.012 (0.31)
<i>Past year return_t</i>	0.044** (2.21)	0.044** (1.89)	0.008 (0.87)	0.042** (2.19)	0.042** (2.17)
Observations	73,964	36,863	36,941	73,964	73,964
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	.639	.656	.582	.639	.640

This panel reports the OLS regression results on differences in high- and low-priced firms' number of off-exchange one-share trades during attention-grabbing events surrounding the introduction of FT. A firm-day observation is classified into the high-priced group if the firm's nominal closing stock price on trading day t equals or exceeds \$100 for columns 1–3 (\$75 for column 4 and \$150 for column 5) and into the low-priced group otherwise. The two subsamples in columns 2 and 3 are divided within high- and low-priced groups based on how a stock's institutional ownership compares to the group median on trading day t . The sample period is from January 2018 to December 2021. *# of one-share trades_{t+1}* is the total number of off-exchange one-share trades on day $t+1$ in thousands. *High price_t* indicates whether the firm is in the high-priced group. *Post-FT_{t+1}* indicates whether trading day $t+1$ falls in the period after FT first became available in the market. The DiD estimator is *High price_t × Post-FT_{t+1}*. Controls are as in Table 2. Detailed variable definitions are in Appendix. Standard errors are clustered by firm and date. All continuous variables are winsorized at the top and bottom 1%. t -statistics are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 5A: Retail herding and tiny trades.

Table 5
(Continued)
B. Reddit discussions

	WSB mention _{t+1}				
	(1) Pooled (\$100)	(2) Low-IO	(3) High-IO	(4) \$75	(5) \$150
<i>High price_t</i>	0.019 (0.12)	0.111 (0.15)	-0.024 (-0.47)	0.002 (0.02)	-0.032 (-0.15)
<i>High price_t × Post-FT_{t+1}</i>	0.352** (2.56)	1.192** (2.17)	0.067 (1.06)	0.286** (2.51)	0.494** (2.21)
<i>ln(Market cap)_{t-1}</i>	0.323*** (3.52)	0.474** (2.46)	0.161*** (2.84)	0.321*** (3.46)	0.331*** (3.69)
<i>Book to market_{t-1}</i>	0.080 (0.40)	0.314 (0.96)	-0.038 (-0.25)	0.078 (0.39)	0.083 (0.41)
<i>Earnings announcement_t</i>	0.057** (2.03)	0.054 (0.81)	0.056*** (2.89)	0.056** (2.03)	0.058** (2.07)
<i>Past month volatility_t</i>	0.068*** (4.96)	0.055*** (2.71)	0.057*** (4.87)	0.068*** (4.98)	0.068*** (4.95)
<i>Past month max return_t</i>	-1.192*** (-2.98)	-1.119* (-1.92)	-0.764*** (-2.69)	-1.198*** (-2.99)	-1.192*** (-2.98)
<i>Past week return_t</i>	0.589*** (4.04)	0.856*** (3.63)	0.090 (1.10)	0.590*** (4.04)	0.589*** (4.04)
<i>Past month return_t</i>	0.015 (0.14)	-0.026 (-0.17)	0.154** (2.26)	0.016 (0.15)	0.015 (0.14)
<i>Past year return_t</i>	0.069* (1.89)	0.043 (0.81)	0.024 (1.04)	0.069* (1.89)	0.069* (1.88)
Observations	73,964	36,863	36,941	73,964	73,964
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	.553	.567	.535	.553	.553

This panel reports the OLS regression results on differences in high- and low-priced firms' number of Reddit discussions during attention-grabbing events surrounding the introduction of FT. A firm-day observation is classified into the high-priced group if the firm's nominal closing stock price on trading day t equals or exceeds \$100 for columns 1–3 (\$75 for column 4 and \$150 for column 5) and into the low-priced group otherwise. The two subsamples in columns 2 and 3 are divided within high- and low-priced groups based on how a stock's institutional ownership compares to the group median on trading day t . The sample period is from January 2018 to December 2021. *WSB mention_{t+1}* is the number of the stock's Reddit discussions on day $t+1$. *High price_t* indicates whether the firm is in the high-priced group. *Post-FT_{t+1}* indicates whether trading day $t+1$ falls in the period after FT first became available in the market. The DiD estimator is *High price_t × Post-FT_{t+1}*. Controls are as in Table 2. Detailed variable definitions are in Appendix. Standard errors are clustered by firm and date. All continuous variables are winsorized at the top and bottom 1%. t -statistics are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 5B: Reddit discussions.

Read: Table 5A shows that attention events produce a larger increase in one-share trades for high-priced stocks post-FT. Table 5B shows a corresponding increase in Reddit discussion.

Economic message: FT-facilitated tiny trades are most powerful when attention coordinates many small investors into the same stocks.

Table 6
FT and price pressure: Price reversals

	BHAR _[t+2,+6]				
	(1) Pooled (\$100)	(2) Low-IO	(3) High-IO	(4) \$75	(5) \$150
<i>High price_t</i>	0.003 (0.96)	0.008 (0.93)	0.006* (1.78)	0.003 (1.03)	-0.006 (-1.42)
<i>High price_t × Post-FT_{t+1}</i>	-0.008*** (-2.71)	-0.016*** (-3.32)	-0.004 (-1.10)	-0.007*** (-2.88)	-0.007* (-1.90)
<i>ln(Market cap)_{t-1}</i>	-0.018*** (-7.99)	-0.018*** (-4.61)	-0.021*** (-8.65)	-0.018*** (-7.91)	-0.018*** (-8.00)
<i>Book to market_{t-1}</i>	0.005 (0.73)	0.002 (0.22)	0.005 (0.76)	0.005 (0.74)	0.005 (0.73)
<i>Earnings announcement_t</i>	0.001 (0.75)	0.001 (0.52)	0.002 (1.11)	0.001 (0.75)	0.001 (0.73)
<i>Past month volatility_t</i>	-0.002*** (-3.02)	-0.002*** (-2.80)	-0.001 (-1.49)	-0.002*** (-3.02)	-0.002*** (-3.01)
<i>Past month max return_t</i>	0.048*** (2.78)	0.050** (2.47)	0.055* (1.90)	0.048*** (2.78)	0.048*** (2.78)
<i>Past week return_t</i>	-0.025*** (-3.66)	-0.026*** (-3.51)	-0.019** (-1.71)	-0.025*** (-3.66)	-0.025*** (-3.65)
<i>Past month return_t</i>	-0.010** (-2.29)	-0.005 (-1.18)	-0.019** (-2.52)	-0.010** (-2.29)	-0.010** (-2.30)
<i>Past year return_t</i>	0.001 (0.75)	-0.000 (-0.40)	0.002** (2.07)	0.001 (0.74)	0.001 (0.77)
Observations	73,964	36,863	36,941	73,964	73,964
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes	Yes
Adjusted R ²	.083	.084	.094	.083	.083

This table reports the OLS regression results on differences in high- and low-priced firms' 5-day buy-and-hold abnormal returns (BHAR) during attention-grabbing events surrounding the introduction of FT. A firm-day observation is classified into the high-priced group if the firm's nominal closing stock price on trading day t equals or exceeds \$100 for columns 1–3 (\$75 for column 4 and \$150 for column 5) and into the low-priced group otherwise. The two subsamples in columns 2 and 3 are divided within high- and low-priced groups based on how a stock's institutional ownership compares to the group median on trading day t . The sample period is from January 2018 to December 2021. *BHAR_[t+2,+6]* is the stock's raw return compounded over $[t+2, t+6]$ minus the corresponding market return compounded over the same window. *High price_t* indicates whether the firm is in the high-priced group on trading day t . *Post-FT_{t+1}* indicates whether trading day $t+1$ falls in the period after FT first became available in the market. The DiD estimator is *High price_t × Post-FT_{t+1}*. Controls are as in Table 2. Detailed variable definitions are in Appendix. Standard errors are clustered by firm and date. All continuous variables are winsorized at the top and bottom 1%. t -statistics are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 6: Subsequent price reversals.

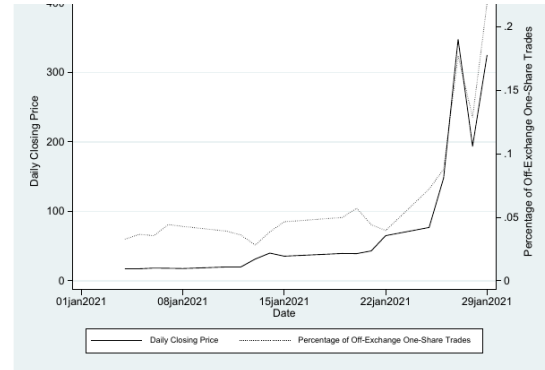


Figure 3
Stock bubbles and percentage of off-exchange one-share trades
This figure plots GameStop's daily closing price (in solid line) and the daily number of off-exchange one-share trades as a percentage of the daily total number of trades (dotted line) during January 2021.

Figure 3: GameStop price and tiny trades.

Read: Table 6 documents lower subsequent abnormal returns after post-FT attention events in high-priced stocks. Figure 3 shows that GameStop’s price spike tracks the share of off-exchange one-share trades during January 2021.

Economic message: The evidence is consistent with temporary price pressure rather than only permanent information discovery.

Table 7
FT and stock bubbles

	Probit(Bubble=1) _{it}					
	(1) Pooled (\$100)	(2) Low-IO	(3) High-IO	(4) \$75	(5) \$150	(6) Alternative Pre
<i>High price_{it}</i>	-2.452*** (-28.77)	-2.598*** (-25.66)	-1.782*** (-17.18)	-2.808*** (-31.93)	-2.449*** (-28.60)	-2.379*** (-13.93)
<i>Post-FT_{it}</i>	0.522*** (4.96)	0.502*** (4.51)	0.813*** (2.95)	0.519*** (4.92)	0.527*** (5.11)	0.351*** (2.72)
<i>High price_{it} × Post-FT_{it}</i>	2.411*** (15.40)	2.463*** (11.92)	1.922*** (5.76)	2.700*** (19.29)	2.344*** (11.50)	2.303*** (10.44)
<i>ln(Market cap)_{it}</i>	-0.204*** (-8.82)	-0.166*** (-6.78)	-0.242*** (-2.72)	-0.199*** (-8.38)	-0.204*** (-9.34)	-0.159*** (-9.45)
<i>Book to market_{q-1}</i>	-0.460*** (-4.73)	-0.531*** (-4.93)	-0.201 (-0.82)	-0.463*** (-4.75)	-0.460*** (-4.78)	-0.435*** (-5.35)
<i>Earnings announcement_{it}</i>	-0.010 (-0.08)	-0.019 (-0.13)	0.024 (0.21)	-0.010 (-0.08)	-0.025 (-0.08)	-0.025 (-0.18)
<i>Past year volatility_{it}</i>	0.017*** (2.69)	0.011** (2.06)	0.056*** (3.97)	0.017*** (2.61)	0.017*** (2.70)	0.031*** (4.69)
<i>Past year max return_{it}</i>	-0.050 (-0.24)	0.023 (0.14)	-1.537*** (-2.83)	-0.040 (-0.19)	-0.050 (-0.24)	-0.308 (-1.52)
<i>Past month return_{it}</i>	0.521*** (3.21)	0.446*** (2.90)	0.654*** (3.59)	0.522*** (3.21)	0.521*** (3.20)	0.390*** (2.87)
<i>Past year return_{it}</i>	-0.014 (-0.56)	-0.019 (-0.87)	0.082 (1.05)	-0.013 (-0.55)	-0.014 (-0.56)	-0.036 (-1.58)
Observations	134,240	67,120	67,120	134,240	134,240	176,298
Pseudo R ²	.188	.162	.175	.188	.188	.147

This table reports the Probit regression results on differences in high- and low-priced firms’ likelihood of experiencing a price bubble surrounding the introduction of FT. Columns 1–5 use firm-month observations from July 2017 to June 2019 as pre-FT period, and column 6 uses firm-month observations from July 1997 to June 1999 as pre-FT period (the dot-com period). The post-FT period is from January 2020 to December 2021. A firm-month observation is classified into the high-priced group if the firm’s nominal stock price at the end of month m equals or exceeds \$100 for columns 1–3 and 6 (\$75 for column 4 and \$150 for column 5) and into the low-priced group otherwise. Columns 1, 4, 5, and 6 use the pooled sample; column 2 uses the subsample of low-IO stocks; and column 3 uses the subsample of high-IO stocks, respectively. The two subsamples in columns 2 and 3 are divided within high- and low-priced groups based on how a stock’s institutional ownership compares to the group median in month m . *Probit(Bubble=1)_{it}* equals one for a firm that experiences a bubble event and zero otherwise. *High price_{it}* whether the firm is in the high-priced group at the end of month m . *Post-FT_{it}* indicates whether month m falls in the post-FT period. The DiD variable is *High price_{it} × Post-FT_{it}*. Controls include the log of market cap of given month (*ln(Market cap)_{it}*), book-to-market of prior quarter (*Book to market_{q-1}*), an indicator for quarterly earnings announcement (*Earnings announcement_{it}*), monthly stock price volatility of past year (*Past year volatility_{it}*), the stock’s maximum monthly return of past year (*Past year max return_{it}*), and the stock’s cumulative return of past month and year (*Past month return_{it}* and *Past year return_{it}*). Detailed variable definitions are in Appendix. Standard errors are clustered by firm and year-month. All continuous variables are winsorized at the top and bottom 1% by year-month. z -statistics are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 7: FT and stock bubbles.

Table 8
FT and stock bubbles: Percentage of tiny trades

	<i>Ret_{it}</i>	
	(1)	(2)
<i># of one-share trades %_{it}</i>	-0.267 (-0.75)	
<i>High price_{it} × # of one-share trades %_{it}</i>	2.501** (2.32)	
<i># of fractional trades %_{it}</i>		-0.406 (-0.58)
<i>High price_{it} × # of fractional trades %_{it}</i>		5.142** (2.39)
<i>ln(Market cap)_{it-1}</i>	-0.028 (-1.10)	-0.265*** (-2.69)
<i>Book to market_{q-1}</i>	0.414** (2.34)	-0.094 (-0.27)
<i>Earnings announcement_{it}</i>	0.004 (0.17)	0.053 (0.91)
<i>Past month volatility_{it}</i>	-0.013*** (-3.80)	-0.025** (-2.67)
<i>Past month max return_{it}</i>	0.225** (2.60)	0.360* (1.71)
<i>Past week return_{it}</i>	-0.055*** (-4.46)	-0.010 (-0.24)
<i>Past month return_{it}</i>	-0.036*** (-3.26)	-0.008 (-0.29)
<i>Past year return_{it}</i>	-0.002 (-1.24)	-0.002 (-0.71)
Observations	2,144	620
Firm fixed effects	Yes	Yes
Date fixed effects	Yes	Yes
Adjusted R ²	.198	.239

This table reports the OLS regression results on differences in high- and low-priced firms’ relation between the daily return and the percentage of the number of tiny trades around bubble events. Column 1 uses the number of off-exchange one-share trades to measure tiny trades and the bubble events are measured from January 2020 to December 2021. Column 2 uses the number of fractional trades detected by Bartlett, McCrary, and O’Hara (2023b) to measure tiny trades and the sample period for the trading day is after March 2021 because of data limitations. A firm-month bubble event is classified into the high-priced group if the firm’s nominal stock price at the end of month m equals or exceeds \$100. For each bubble event, we include 5 trading days before and after the peak day in the regression analysis. *Ret_{it}* is the firm’s raw stock return of trading day t . *# of one-share trades %_{it}* is the daily number of off-exchange one-share trades as a percentage of total trades on trading day t . *# of fractional trades %_{it}* is the daily number of fractional trades as a percentage of daily total trades on trading day t . *High price_{it}* denotes whether the firm is in the high-priced group at the end of month m . Controls include the log of market cap on the previous trading day (*ln(Market cap)_{it-1}*), book-to-market of prior quarter (*Book to market_{q-1}*), an indicator for quarterly earnings announcement (*Earnings announcement_{it}*), stock price volatility of past month (*Past month volatility_{it}*), the stock’s maximum daily return of past month (*Past month max return_{it}*), and the stock’s cumulative return of past week, month, and year (*Past week return_{it}*, *Past month return_{it}*, and *Past year return_{it}*) as well as firm and date fixed effects. Detailed variable definitions are in Appendix. Standard errors are clustered by firm and date. All continuous variables are winsorized at the top and bottom 1%. t -statistics are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 8: Percentage of tiny trades during bubbles.

Read: Table 7 shows that high-priced stocks become more likely to experience bubbles after FT. Table 8 shows that tiny-trade intensity is more closely associated with daily returns for high-priced bubble stocks.

Economic message: FT can make high nominal-price stocks more exposed to meme-stock-like bubbles by allowing small investors to participate in dollar amounts.

Table 9
FT and stock bubbles: Feedback effect

	<i>Probit(Bubble=1)_{it}</i>		
	(1) WSB posts	(2) Credit rating	(3) Feedback
<i>High price_{it}</i>	-2.513*** (-20.29)	-2.545*** (-8.29)	-2.441*** (-20.96)
<i>Post-FT_{it}</i>	0.473*** (3.75)	0.197 (0.72)	0.500*** (3.61)
<i>Subsample indicator</i>	0.313** (2.06)	-0.796* (-1.79)	0.050 (0.36)
<i>High price_{it} × Post-FT_{it}</i>	-0.478*** (-3.40)	-0.272 (-0.80)	2.000*** (8.97)
<i>High price_{it} × Subsample indicator</i>	-0.029 (-0.17)	0.415 (0.97)	-0.057 (-0.36)
<i>Post-FT_{it} × Subsample indicator</i>	0.131 (0.84)	0.145 (0.33)	-0.062 (-0.40)
<i>High price_{it} × Post-FT_{it} × Subsample indicator</i>	3.039*** (13.57)	3.093*** (7.22)	0.751*** (2.71)
<i>lnMarket cap_{it}</i>	-0.243*** (-10.34)	-0.240** (-2.34)	-0.204*** (-8.39)
<i>Book to market_{q-1}</i>	-0.412*** (-4.15)	0.170 (0.70)	-0.471*** (-4.26)
<i>Earnings announcement_{it}</i>	-0.025 (-0.20)	-0.067 (-0.61)	-0.007 (-0.06)
<i>Past year volatility_{it}</i>	0.014** (2.09)	0.055** (2.41)	0.026*** (3.85)
<i>Past year max return_{it}</i>	-0.017 (-0.08)	-0.572 (-0.69)	-0.297 (-1.30)
<i>Past month return_{it}</i>	0.526*** (3.22)	0.574* (1.78)	0.573*** (3.28)
<i>Past year return_{it}</i>	-0.024 (-0.95)	0.033 (0.43)	-0.007 (-0.29)
Observations	128,761	45,969	130,835
Pseudo R ²	.208	.328	.210

This table reports cross-sectional analyses of the Probit regression results on differences in high- and low-priced firms' likelihood of experiencing a price bubble surrounding the introduction of FT. The pre-FT period is from July 2017 to June 2019 and the post-FT period is from January 2020 to December 2021. Column 1 uses the pooled sample in Table 7 excluding stocks with tickers that have special meanings. *Subsample indicator* in column 1 indicates whether the total number of times of a stock being discussed on the forum from January 2020 to December 2021 equals or exceeds the sample median. Column 2 uses the pooled sample in Table 7 with credit rating data. *Subsample indicator* in column 2 indicates whether the credit rating is of BB+ or lower. Column 3 uses the pooled sample in Table 7 with available capital-to-price sensitivity estimated over the past eight quarters. *Subsample indicator* in column 3 indicates whether the firm exhibits a capital-to-price sensitivity that equals or exceeds the sample median. *Probit(Bubble=1)_{it}* equals one if a firm that experiences a bubble event and zero otherwise. *High price_{it}* denotes whether the firm is in the high-priced group at the end of month *m*. *Post-FT_{it}* indicates whether month *m* falls in the post-FT period. The key DiD variable is the three-way interaction of *High price_{it}*, *Post-FT_{it}*, and *Subsample indicator*. Controls include the log of market cap of the given month (*lnMarket cap_{it}*), book-to-market of prior quarter (*Book to market_{q-1}*), an indicator for quarterly earnings announcement (*Earnings announcement_{it}*), monthly stock price volatility of past year (*Past year volatility_{it}*), the stock's maximum monthly return of past year (*Past year max return_{it}*), and the stock's cumulative return of past month and year (*Past month return_{it}* and *Past year return_{it}*). Detailed variable definitions are in Appendix. Standard errors are clustered by firm and year-month. All continuous variables are winsorized at the top and bottom 1% by year-month. z-statistics are in parentheses. **p* < .1; ***p* < .05; ****p* < .01.

Table 9: Feedback-effect tests.

Appendix

Table A1
Variable definitions

Variable	Definition
<i>Variables used in baseline analyses</i>	
<i>Post-IB-FID_{it}</i>	An indicator that equals one if day <i>t</i> falls between November 25, 2019, when Interactive Brokers introduced FT and January 28, 2020, the day before Fidelity introduced FT, and zero otherwise. This indicator captures when FT was available through either Interactive Brokers or Robinhood.
<i>Post-FID_{it}</i>	An indicator that equals one if day <i>t</i> falls on or after January 29, 2020, when Fidelity introduced FT. This indicator captures when FT was available through all three brokers.
<i>High price</i>	An indicator that equals one if the stock price as the end of November 2019, when Interactive Brokers first introduced FT, equals or exceeds \$100 (alternatively, \$75 or \$150 in robustness checks) and zero otherwise.
<i># of one-share trades_{it}</i>	The total number of off-exchange one-share trades detected from TAQ on day <i>t</i> in thousands.
<i>lnMarket cap_{it-1}</i>	Natural logarithm of market capitalization ($\text{PRC} \times \text{SHROUT}$) on trading day <i>t</i> - 1.
<i>Book to market_{q-1}</i>	Book value of assets (ATQ) divided by market value of assets ($\text{PRCCQ} \times \text{CSHOQ} + \text{LTQ}$) at the end of quarter <i>q</i> - 1.
<i>Earnings announcement_{it}</i>	An indicator that equals one if day <i>t</i> falls within a 3 day window centered on a quarterly earnings announcement and zero otherwise.
<i>Past month volatility_{it}</i>	The standard deviation of the stock's daily returns over the past 30 calendar days in percentage points.
<i>Past month max return_{it}</i>	The stock's maximum daily return over the past 30 calendar days, calculated following the method of Bali, Cakici, and Whitelaw (2011).
<i>Past week return_{it}</i>	The stock's cumulative return of the past week by compounding the stock's daily returns over the past 7 calendar days.
<i>Past month return_{it}</i>	The stock's cumulative return of the past month by compounding the stock's daily returns over the past 30 calendar days.
<i>Past year return_{it}</i>	The stock's cumulative return of the past year by compounding the stock's daily returns over the past 360 calendar days.
<i>Additional variables used in identification analyses</i>	
<i>Post-ZCT-FT_{it}</i>	An indicator that equals one if day <i>t</i> falls between October 1, 2019, when Charles Schwab introduced ZCT and November 24, 2019, the day before Interactive Brokers introduced FT, and zero otherwise. This indicator captures the period when ZCT was available but FT was not.
<i>Post-FID-COVID_{it}</i>	An indicator that equals one if day <i>t</i> falls between January 29, 2020, when Fidelity introduced FT and February 29, 2020, when Washington declared the state of emergency, and zero otherwise. This indicator captures the period after FT became widely available but before COVID-19 lockdowns.
<i>Post-COVID_{it}</i>	An indicator that equals one if day <i>t</i> falls on or after March 1, 2020, the first day of the month when all states declared emergency, and zero otherwise.
<i>Post-RH_{it}</i>	An indicator, used only in Robinhood-specific analyses, that equals one if day <i>t</i> falls on or after December 12, 2019, when the broker introduced FT and zero otherwise.
<i>Post-CS_{it}</i>	An indicator, used only in Charles Schwab-specific analyses, that equals one if day <i>t</i> falls on or after June 9, 2020, when the broker began FT for the S&P 500 firms and zero otherwise.
<i>High price_{CS}</i>	An indicator that equals one if the stock price as of June 8, 2020, the day before Charles Schwab began FT for the S&P 500 firms, equals or exceeds \$100 and zero otherwise.
<i>S&P 500</i>	An indicator that equals one if the firm is included in the S&P 500 index and zero otherwise.
<i>RH trading intensity_{it}</i>	The sum of the absolute value of intraday hourly changes in the number of Robinhood users holding the stock on trading day <i>t</i> scaled by the number of Robinhood users holding the stock at the end of the previous trading day.

Table A1: Variable definitions, part 1.

Read: Table 9 tests whether the FT-bubble relation is stronger in settings where feedback effects should be larger. Table A1 defines the baseline and identification variables.

Economic message: The bubble evidence is strongest when social-media coordination or financing feedback can amplify retail price pressure.

Table A1 (Continued)	
Variable	Definition
Additional variables used in price pressure analyses	
$Post-FT_{t+1}$	An indicator that equals one if day $t+1$ falls on and after November 25, 2019, when Interactive Brokers introduced FT and zero otherwise. This indicator captures when FT was introduced to the market
$High\ price_t$	An indicator that equals one if the stock price at the end of trading day t equals or exceeds \$100 (alternatively, \$75 or \$150 in robustness checks) and zero otherwise
$\# of\ one\text{-}share\ trades_{t+1}$	The total number of off-exchange one-share trades detected from TAQ on day $t+1$ in thousands
$WSB\ mention_{t+1}$	The number of the stock's Reddit discussions on day $t+1$
$BHAR_{[t+2, t+6]}$	Buy and hold abnormal return over the next 5 trading days. It is computed as the firm's raw daily return compounded from day $t+2$ to day $t+6$ minus the corresponding daily return on the CRSP value-weighted index compounded over the same window, $\prod_{t=t+2}^{t+6}(1+r_t) - \prod_{t=t+2}^{t+6}(1+r_{mt})$
Additional variables used in stock bubble analyses	
$Prob(Bubble=1)$	An indicator that equals one if the stock experiences a bubble in the next 6 months and zero otherwise. We define bubble occurrence as when the stock's peak price during the first 3 months (i.e., $m+1$ to $m+n$ with $0 \leq n \leq 3$) is more than 150% of its price at the end of month m but the stock's subsequent trough price during the following 3 months (i.e., $m+n$ to $m+n+3$) drops at least 40% from the peak price
Ret_t	The stock's daily raw return on trading day t
$Post-FT_m$	An indicator that equals one if month m falls in or after February 2020. This indicator captures the period when FT is widely available in the market
$High\ price_m$	An indicator that equals one if the stock price at the end of month m equals or exceeds \$100 (alternatively, \$75 or \$150 in robustness checks) and zero otherwise
<i>Subsample indicator</i>	An indicator that equals one if the firm's total number of times being discussed on the Reddit WSB forum from January 2020 to December 2021 equals or exceeds the sample median and zero otherwise in Table 9, column 1. An indicator that equals one if the firm carries a credit rating of BB+ or lower and zero otherwise in Table 9, column 2. An indicator that equals one if the firm exhibits a high capital-to-price sensitivity over the past eight quarters equal to or exceeding the sample median and zero otherwise in Table 9, column 3
$\# of\ one\text{-}share\ trades\ \%_t$	The daily number of off-exchange one-share trades as a percentage of total number of trades on trading day t
$\# of\ fractional\ trades\ \%_t$	The daily number of fractional trades detected by Bartlett, McCrory, and O'Hara (2023b) as a percentage of total number of trades on trading day t
$\ln Market\ cap_m$	Natural logarithm of market capitalization ($\frac{PRC}{SHROUT}$) at the end of month m
$Book\ to\ market_{q-1}$	Book value of assets (ATQ) divided by market value of assets ($\frac{PRCCQ \times CSHOQ + LTQ}{}$) at the end of quarter $q-1$
$Earnings\ announcement_m$	An indicator that equals one if month m has a quarterly earnings announcement and zero otherwise
$Past\ year\ volatility_m$	The standard deviation of the stock's monthly returns over the past 12 months in percentage points
$Past\ year\ max\ return_m$	The stock's maximum monthly return over the past 12 months
$Past\ month\ return_m$	The stock's cumulative return of past month by compounding the stock's daily returns during the month
$Past\ year\ return_m$	The stock's cumulative return of past year by compounding the stock's monthly returns over the past 12 months
This appendix describes the calculation of variables used in the main analyses. Underlined variables refer to variable names within Compustat or CRSP. t indexes trading day, m indexes month, q indexes quarter, and y indexes year, respectively.	

Table A1: Variable definitions, part 2.

Read: The final appendix panel defines the price-pressure and stock-bubble variables used in the event and bubble tests.

Economic message: The paper's interpretation depends on carefully separating tiny-trade activity, social-media attention, reversals, bubble definitions, and feedback-effect proxies.